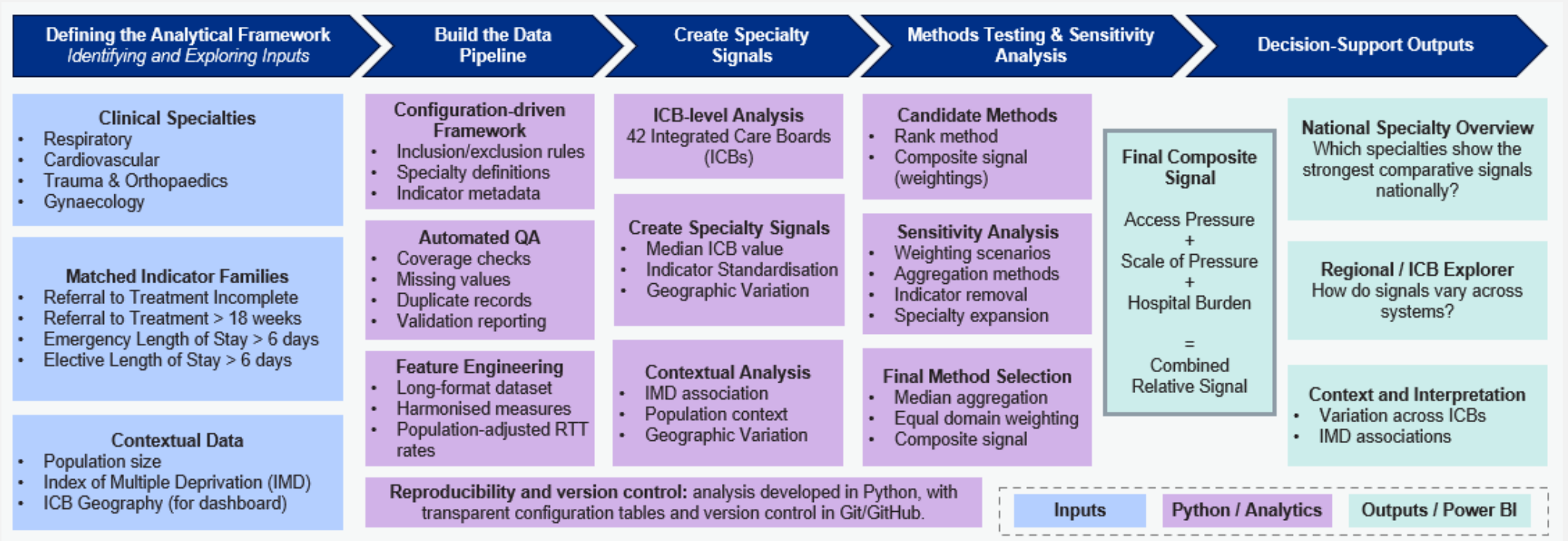
Identifying Signals of Unmet Need Using NHS System Data

A reproducible framework for specialty-level screening and prioritisation

(1) The Problem and Context

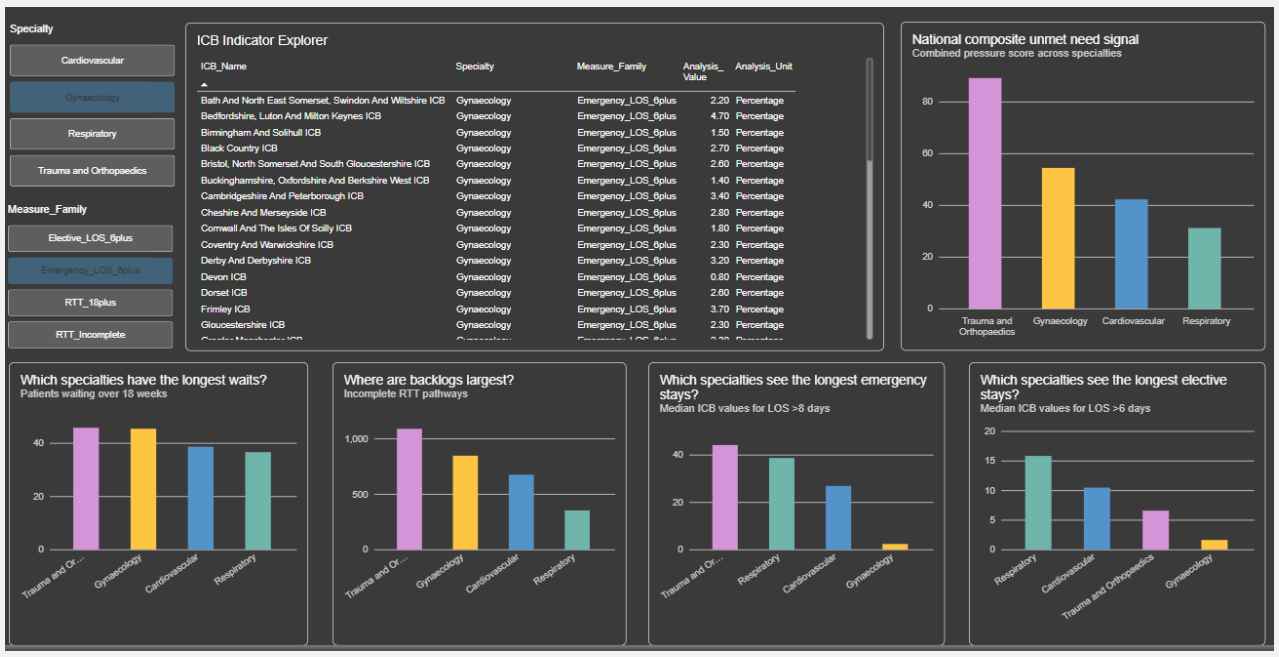
Identifying areas of greatest unmet need for research and innovation (R&I) investment currently relies on a combination of policy priorities and clinical expertise. There is an opportunity to explore whether NHS system data could provide an additional evidence layer to support prioritisation. This project tested whether a reproducible, data-driven framework could be developed to identify specialties with potential signals of unmet need, using indicators of system pressure, hospital burden and and geographical variation.

(2) Data and Methodology: A reproducible Python workflow was developed to clean and harmonise selected NHS indicators, generate comparable specialty-level signals, test the robustness of results, and produce national and ICB-level Power BI outputs.



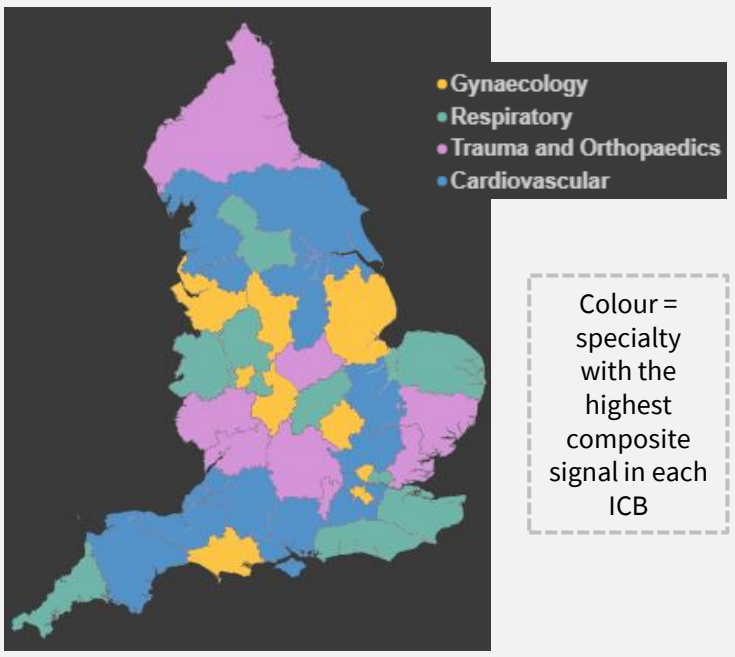
Proof of concept: four matched indicators across four specialties and 42 ICBs, using 2024/25 data.

Figure 1: National view: comparative specialty signals



The national dashboard provides a comparative overview across specialties, helping identify areas that may warrant further investigation. In this proof of concept, Trauma & Orthopaedics demonstrated the strongest overall signal, with the ordering remaining broadly stable across alternative weighting and sensitivity-testing scenarios.

Figure 2: Regional view: leading specialty signal by ICB



The regional map highlights how the leading specialty signal varies between ICBs. This demonstrates the value of considering both national and regional perspectives when exploring areas of unmet need.

(3) Key Learnings

- Data quality, availability and consistency varied substantially across indicators / specialties
- Building comparable indicator families provide more challenging than expected
- ICB geography changes in 2026 required careful alignment of datasets and selection of a consistent reporting period (2024/25)
- Population and deprivation were more useful as contextual than scoring variables
- Different specialties require different measures (e.g. outpatient pathways)

(4) Business Value and Impact

Demonstrates how routinely available NHS system data can provide a transparent and reproducible screening layer to complement clinical expertise and policy priorities.

Potential benefits include:

- more systematic and transparent prioritisation
- improved reproducibility and auditability
- a scalable framework that can be expanded across specialties and indicators

(5) Next Steps: clinical validation; additional specialties; dashboard refinement

Future Development: time-series trends; validation of indicator selection and weighting; expansion to additional specialties; enhanced inequalities analysis.

Long-term Ambition: a scalable and reproducible evidence layer to support NHS research and innovation prioritisation.

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Open repository: code, methods & outputs